Question 4

The body produces a hormone known as stophin in response to consumption of a compound known as vitrogo. Stophin is only produced in response to vitrogo so a person who does not take vitrogo has no stophin in his blood. For 6 persons given different amounts of Vitrogo, the following amounts of Stophin are observed in their blood:

X= Amount of Vitrogo	1	2	3	4	5	6
Y = Amount of Stophin	2	2	6	12	20	18

A. Dr. Z fits the following linear model with no intercept to the data:

$$Y = bX + \epsilon \text{ where } E[\epsilon] = 0 \text{ and } Var[\epsilon] = \sigma^2$$

Find the Least Squares estimate $\hat{b}$ for b and estimate the variance of $\hat{b}$

B. Dr. W agrees with the linear model of Dr. Z with no respect to no intercept but believes that the variance of Y given X is proportional to X^2, namely

$$Y = bX + \epsilon \text{ where } E[\epsilon] = 0 \text{ and } Var[\epsilon] = X^2\sigma^2$$

If we let W = Y/X this is the same as

$$W = b + \epsilon' \text{ where } E[\epsilon'] = 0 \text{ and } Var[\epsilon'] = \sigma^2$$

Find the Least Squares estimate $\hat{\hat{b}}$ for b using this model

C. Compare $\hat{b}$ to $\hat{\hat{b}}$ and looking at the individual values of Y and X, explain why these estimates differed in the direction observed.